

Technology Stack

Date	5 July 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular; HTML/CSS)	React.js (with Vite), HTML5, CSS3, JavaScript (ES6+)	React.js provides a fast, component-based UI for the dashboard, loans, and settlement pages. Vite gives fast builds and hot reload during development.
2	Backend / Server-Side (e.g., Node.js, Python; Spring Boot)	Python, FastAPI	FastAPI is a modern, high-performance Python framework ideal for building the RESTful APIs behind authentication, loan management, and AI-driven features, with automatic validation and docs.
3	Database (e.g., MySQL, PostgreSQL; MongoDB)	SQLite	SQLite is lightweight, serverless, and easy to set up — ideal for storing Users, Loans, Financial Profiles, Settlement Records, and AI History without extra infrastructure.
4	ORM / Database Toolkit (e.g., SQLAlchemy, Prisma)	SQLAlchemy ORM	SQLAlchemy ORM provides powerful, flexible database interaction with Python, making it easy to define and query the five core models matching the ER diagram.

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5	AI / ML Services (e.g., OpenAI API, Google Gemini API)	Google Gemini API	Gemini API is used for AI-driven settlement negotiation strategy and letter generation, with a rule-based fallback engine ensuring the feature works even if the API is unavailable.
6	API Communication (e.g., REST, GraphQL)	RESTful APIs	RESTful APIs are used for all communication between the React frontend and FastAPI backend, keeping the system simple, scalable, and easy to test.
7	Development Tools (e.g., VS Code, Git)	VS Code, Git, GitHub	VS Code is used as the IDE. Git and GitHub are used for version control, collaboration across the 5-person team, and code management.
8	Deployment (e.g., Vercel, Render, AWS, Docker)	Render (Backend), Vercel (Frontend)	Render is used to deploy the FastAPI backend, and Vercel is used to deploy the React frontend — both offering fast, reliable, free-tier hosting suitable for this project.